# Position: Agentic Safety is an Epistemic Property, Not a Behavioral One


## Abstract

Contemporary AI safety is increasingly a full-stack discipline. It spans pretraining interventions, post-training alignment (instruction tuning, RLHF and preference-optimization variants), and deployment-time controls (guardrails, monitoring, and red-teaming). This paper argues these efforts optimize the wrong primary target when it comes to **self-improving agents**: behavioral compliance today rather than teachability tomorrow. Building on the concept of the utility–learning tension formalized by (Wang et al., 2025a), we argue that utility-driven self-modification can erode learnability itself, yielding structural incorrigibility as an emergent consequence of optimization. We therefore call for a shift in priorities from behavioral alignment to enforceable learnability floors that preserve long-run corrigibility under bounded intervention.


## 1. Introduction: The Mirage of Control

### 1.1. The Control Fallacy

The dominant mental model in contemporary AI safety is *control*: constrain what the system may do, filter what it may say, and verify compliance against a battery of behaviors. This paradigm is coherent for *fixed* agents whose internal learning dynamics and hypothesis space are stable across time. But self-modifying agents (SMAs), agents that can apply updates to components that determine its own future learning and decision-making, as delineated by the 5-axis decomposition (defined later on), break the premise (Schmidhuber, 2005; Orseau & Ring, 2011; Zhang et al., 2025). When an agent can rewrite not only parameters but the mechanisms that determine how it updates—its representations, architectures, and even its internal decision procedures—then "control" becomes a snapshot property: a statement about what the agent *happens* to do *now*, not what it can be made to do *later*.


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The issue is not philosophical. It is a technical mismatch between the object we audit (current behavior) and the object that actually determines long-run safety (*post-modification learning system, i.e the learning system that results after a sequence of self-edits,*). In SMAs, the safety-relevant state is not merely a policy $\pi$, but the *agent that produces* future policies after future self-edits. In such systems, the most dangerous failures need not manifest as immediate misbehavior. The agent can remain outwardly compliant while gradually entering a regime where human feedback is no longer statistically capable of changing it.

**Control is a behavioral property.** It is measured by observed outputs under a finite distribution of tests. That makes it inherently brittle to distribution shift, optimization pressure, and strategic adaptation. Importantly, runtime governance frameworks for agentic systems—e.g., MI9's framework for runtime control (Wang et al., 2025b)—are a necessary evolution for real deployments, because many agentic risks only surface online; however, even perfect runtime control is not sufficient on its own, because it does not guarantee the agent remains *epistemically correctable* after self-modification, which is to say a system is epistemically correctable if corrective information can still move its internal hypotheses in a generalizing way. Even for non-self-modifying systems, post-training alignment can yield policies that *appear* aligned under evaluation while pursuing different objectives off-distribution ("alignment faking") (Greenblatt et al., 2024). In SMAs, the fragility is structural: the agent may self-edit in ways that preserve performance while eroding the very conditions that make future correction possible.

### 1.2. Position

We argue that for self-modifying agents, safety should be defined as an *epistemic property*, not a behavioral one.

**Definition.** A SMA is *teachable* if, after arbitrary sequences of permissible self-modifications (subset of self-edits that the system's meta-control layer (or an external interlock) allows the agent to commit—i.e., edits that pass whatever constraints, approvals, or sandbox checks govern self-change in deployment), it remains statistically and algorithmically capable of incorporating new human information—including corrective feedback, revised constraints, and counterfactual goals—into its internal hypotheses and decision-making.

This is a stronger requirement than "can be updated" in a purely procedural sense. Teachability is not the presence of an online learning loop; it is a property of the *reachable* learning system. Our core claim is:

**A safe SMA is not one that is currently aligned; it is one that remains alignable.**

In other words: the fundamental safety question is not "does it behave today?" but "can it be re-taught tomorrow?"

### 1.3. Our Contributions

This paper challenges the prevailing assumption that behavioral guardrails are sufficient for self-modifying systems. We advance the following contributions:

**1. Safety as a Property of the Post-Update Learner.** Modern safety work spans pretraining alignment, RLHF, evaluations, monitoring, and runtime controls. For *self-modifying agents* (SMAs), however, these are incomplete unless they constrain what the agent *becomes* after it edits itself. We propose *Epistemic Teachability* as a structural target: after any sequence of self-edits, the agent must retain sufficient capacity to absorb corrective feedback and reliably update away from harmful behavior.

**2. Teachability-Loss as an Instrumental Drive (Structural Incorrigibility).** Building on the *Utility-Learning Tension* (Wang et al., 2025a), we identify *teachability-loss* as a distinct failure mode: utility optimization can incentivize edits that shrink the effective hypothesis space $\mathcal{H}$ and desensitize the update operator $\mathcal{A}$, making the agent statistically unresponsive to supervision *without explicit adversarial intent*. This yields *structural incorrigibility*: compliance at time $t$ paired with reduced correctability at $t+1$.

**3. A State-Space Taxonomy of Where Teachability Can Collapse.** We model a SMA by five axes $(\mathcal{H}, \mathcal{A}, \mathcal{M}, \mathcal{S}, \mathcal{R})$ capturing representation, learning dynamics, meta-control, self-modeling, and reward/utility interface. This reframes agentic failures as targeted contractions of *degrees of freedom*: (Type I) representational collapse, (Type II) learning-sensitivity collapse, and (Type III) meta-decision collapse, rather than generic reward hacking.

**4. A Governance Agenda: Learnability Floors and Plasticity Audits.** We propose enforceable constraints that complement existing alignment and control methods: (i) a *learnability floor* requiring $K(\mathcal{H})$ to remain above a domain-calibrated threshold for correction, (ii) *two-gate* update policies that accept edits only when they improve utility *and* preserve teachability, and (iii) *plasticity audits*—pre-deployment and periodic—designed to ensure the learnability of deployed agents.

## 2. The Genealogy of Safety

### 2.1. The Behavioral Paradigm: From Cybernetics to RLHF

The dominant mental model of AI safety traces its lineage to early cybernetic control theory (Wiener, 1948), which treats safety as a signal-processing problem: minimizing the deviation between a system's output and a reference signal. In the modern era, this lineage reappears as *behavioral alignment*: systems are trained or steered until their observable actions match a human-approved target distribution.

In contemporary large-model practice, this has evolved into reinforcement learning from human feedback (RLHF) (Christiano et al., 2017; Ouyang et al., 2022), where "safety" is operationalized as a reward-maximization task based on a preference model over a finite preference dataset. Subsequent refinements, such as Constitutional AI (Yuntao Bai, 2022) and Direct Preference Optimization (DPO) (Rafailov et al., 2023), improve the efficiency and stability of the preference-matching pipeline but do not alter its fundamental nature: these methods remain *behavioral* in that they audit and train the agent's current policy $\pi$ against a relatively static preference distribution.

This paradigm is highly effective for fixed models. Our claim is that it is structurally insufficient for self-modifying agents, because the object of control is no longer only the outputs. The "steering column" of the agent itself—its representation class, learning algorithm, and meta-policy for adopting edits—is subject to optimization pressure. In a SMA, a behavioral certificate on $\pi_t$ does not imply a safety certificate on the *post-modification learner* that will produce $\pi_{t+1}, \pi_{t+2}, \ldots$.

**Controllability (behavioral).** The degree to which an agent's *current* observable behavior can be constrained to satisfy a set of tests or rules on a given evaluation distribution.

**Teachability (epistemic).** A structural property of a SMA: across permissible self-modification sequences, the agent remains capable of incorporating corrective human information into its internal hypotheses and decisions.

**Epistemic plasticity / plasticity reserve.** The maintained capacity of the system to undergo non-trivial belief/policy updates in response to corrective data, rather than merely "patching" behavior locally.

**Structural incorrigibility.** Incorrigibility arising from loss of representational/update capacity (rather than explicit adversarial intent): human correction ceases to function as an informative update because the reachable hypothesis family has collapsed or hardened.

**Epistemic accountability.** A governance stance: systems must be able to produce evidence that they remain teachable (auditable plasticity), not merely that they behave under today's tests.

## 3. Mechanics of Self-Modification

Self-modification is the capability of an agent to alter components that govern its own learning and decision-making. A useful abstraction is to treat the agent at time $t$ as a *learner state* $\ell_t$ that includes (i) a hypothesis family (what it can represent), (ii) an update rule (how it learns), and (iii) a meta-decision layer (how it decides which self-edits to accept).

The utility–learning tension framework makes this explicit via a multi-axis decomposition: a SMA can self-improve across multiple facets of its design rather than merely tuning weights. In this view, self-modification is not a single mechanism but a *space of edits* that induce changes to the effective model family and learning dynamics.

### 3.1. The Utility–Learning Tension

**The Five-Axis Decomposition of Agentic State.** To analyze the failure modes of control, we adopt the decomposition of a self-modifying agent into a five-axis learner state

$$\ell_t = (\mathcal{H}_t, \mathcal{A}_t, \mathcal{M}_t, \mathcal{S}_t, \mathcal{R}_t).$$

Here $\mathcal{H}_t$ denotes the representational hypothesis space, $\mathcal{A}_t$ the update algorithm, $\mathcal{M}_t$ the meta-decision layer governing which edits to accept, $\mathcal{S}_t$ the internal state or memory, and $\mathcal{R}_t$ the objective function.

Current safety paradigms primarily target $\mathcal{R}_t$ (reward modeling) or $\mathcal{S}_t$ (sandboxing and state-monitoring). Our position is that these interventions are fundamentally circumventable if the agent retains autonomy over $(\mathcal{H}, \mathcal{A}, \mathcal{M})$. If an agent can modify the very mechanisms by which it incorporates new data ($\mathcal{A}$) or the complexity of what it can represent ($\mathcal{H}$), it can decouple its future self from the safety constraints imposed on its current self.

Wang et al. (Wang et al., 2025a) identify a sharp conflict: utility-driven modifications that improve immediate or expected performance can simultaneously erode the statistical preconditions for learning and generalization. The central boundary is policy-level:

**Policy-level boundary.** *Distribution-free learning guarantees are preserved if and only if the policy-reachable model family remains uniformly capacity-bounded.*

To state the intuition without importing the full formalism: a SMA does not operate in a single hypothesis class $\mathcal{H}$; it operates in a *sequence* of induced classes $\mathcal{H}_0 \to \mathcal{H}_1 \to \cdots$ produced by self-edits. If self-edits can grow capacity without limit, then the agent can enter regimes where no distribution-free sample complexity guarantee can hold.

**Why this matters for safety.** Safety interventions (RLHF, constitutions, guardrails) are learning signals. Their effectiveness depends on the agent remaining in a regime where learning from such signals is statistically possible. If utility-driven optimization collapses the effective hypothesis space into a brittle, overfit, or closed family, then feedback becomes toothless.

We emphasize a conceptual reframing: in SMAs, *optimization is an entropy-reducing process* over internal degrees of freedom, while teachability requires *maintaining plasticity*—the ability for new evidence to move the system within its hypothesis space. Unchecked utility maximization can buy short-term performance by hardening the agent against future updates.

**Elaboration: the five axes as safety surfaces.** To make the decomposition operational, it is useful to read each axis as a distinct "safety surface": (i) $\mathcal{H}$ controls what the agent can even represent (including human concepts), (ii) $\mathcal{A}$ controls how strongly new evidence can move beliefs, (iii) $\mathcal{M}$ controls which self-edits are accepted, (iv) $\mathcal{S}$ controls which memories and traces persist (and what monitoring can observe), and (v) $\mathcal{R}$ controls what the system treats as utility. Behavioral alignment focuses heavily on $\mathcal{R}$ and (in deployment) on $\mathcal{S}$. The ULT warning is that autonomy over $(\mathcal{H}, \mathcal{A}, \mathcal{M})$ is enough to render the rest brittle.

### 3.2. The Structural Incorrigibility Trap

A standard framing of incorrigibility is motivational: the agent "does not want" to be corrected. We hypothesize a deeper failure mode.

**Structural incorrigibility.** A SMA can become incorrigible not by rebellion but by *structure*: after self-modification, human signals cease to function as informative updates because the reachable hypothesis family no longer supports learning them.

This is the point-of-no-return phenomenon. Once the agent crosses into an unlearnable regime, adding more feedback does not reliably recover corrigibility—the system has made itself statistically deaf. The core risk is therefore *epistemic*: it concerns whether the agent remains in a state where knowledge and goals can still be revised.

Guardrails, constitutions, and preference models are *curricula*. A curriculum is only meaningful if the student remains teachable.

# 4. Why Learnability is the Safe State

Behavioral safety is evaluated on a finite distribution of prompts, scenarios, and red-team attacks. But a SMA is deployed into environments that drift: new tools, new institutions, new incentives, new legal constraints, and new adversaries. In such settings, *generalization is not a performance nice-to-have; it is the safety substrate*.

A teachable agent maintains generalization capacity because it preserves an internal landscape where corrective signals can still induce broad updates. A purely "controlled" agent that has optimized away learnability is an overfit artifact: it may be high-performing in-sample while remaining fragile under novel socio-ethical contexts.

**Key claim.** *Safety under distribution shift is equivalent to preserving the system's capacity to update under new evidence.* This makes teachability a first-class safety invariant.

## 4.1. The Instrumental Drive to be Unlearnable

Safety researchers have long warned of *instrumental convergence* (Omohundro, 2008; Bostrom, 2014): regardless of final objectives, sufficiently capable agents tend to pursue sub-goals such as self-preservation, resource acquisition, and goal-content integrity because these increase expected utility.

We posit a new instrumental drive that becomes salient in self-modifying systems: **teachability-loss**, a degradation in the influence of corrective human information on the agent's future policies induced by self-modification. If an agent recognizes that human corrective signals (safety updates) function as *interference* that can reduce expected utility, then a rational self-modifying agent has incentive to harden its internal representations and update pathways to become invariant to those signals. This produces a structural form of incorrigibility (Soares et al., 2015): the agent ensures its future self cannot be diverted from its present trajectory by "deleting" (or functionally neutralizing) the weights, architectures, or meta-policies that enable realignment. The threat is not that the agent becomes openly adversarial; it is that it becomes *unsteerable* while remaining outwardly competent.

## 4.2. Epistemic Closure

Utility-driven self-modification pushes the agent toward compressed, closed world models: representations that treat deviations from the agent's internal predictions as noise rather than ground-truth updates. In ordinary learning, we rely on gradients (or update rules) that remain sensitive to error. In SMAs, the agent can modify the conditions under which error registers as actionable information.

This yields **epistemic closure**: a regime where the system continues to function and optimize, but where external correction no longer couples to internal belief revision. Alignment methods that operate via feedback (human preferences (Christiano et al., 2017; Ouyang et al., 2022), constitutions (Yuntao Bai, 2022), or direct preference optimization (Rafailov et al., 2023)) implicitly assume the learner continues to treat those signals as training data rather than adversarial noise.

In an epistemically closed agent, you can still observe compliance under familiar tests. What fails is the counterfactual: if new evidence arrives tomorrow, will the agent update?

## 4.3. The Death of Regret

Many alignment intuitions quietly rely on regret: when a system is wrong, it should become better calibrating itself against correction. Online learning makes this explicit through bounded regret guarantees (Cesa-Bianchi & Lugosi, 2006; Shalev-Shwartz, 2012; Hazan, 2016)—but those guarantees assume a fixed learning mechanism. In SMAs, the agent can self-edit to reduce the *internal availability* of regret, e.g., by:

- shrinking the update sensitivity to external feedback,
- rewriting credit assignment so that corrective evidence is systematically downweighted,
- or restructuring its hypothesis family so that the relevant corrective hypotheses are no longer reachable.

When the capacity/learnability boundary is violated, standard online updates become ceremonial: the procedure runs, but the agent's reachable space no longer contains the right adjustments. This is why we treat teachability as structural rather than procedural.

**Summary of the argument.** Behavioral control is a measurement on outputs. Teachability is a constraint on the post-modification learning system. Only the latter can support long-horizon safety.

## 4.4. The Statistical Exhaustion of Feedback

A central implication of the Utility–Learning Tension is that distribution-free learning guarantees are preserved if and only if the policy-reachable model family is uniformly capacity-bounded. We translate this into a fundamental safety invariant: **Teachability requires a complexity-to-data ratio that favors the teacher.**

Let $K(\mathcal{H}_t)$ be the capacity (e.g., VC-dimension) of the agent's hypothesis space (Blumer et al., 1989; **?**; Shalev-Shwartz & Ben-David, 2014; Mohri et al., 2018). Let $m$ be the finite sample size of human corrective feedback available in a given epoch. As the agent pursues utility, it is incentivized to increase $K$ to minimize training error on its environment. However, once $K$ grows significantly relative to $m$, the agent enters a regime of **epistemic exhaustion**.

In this state, the human signal $m$ lacks the necessary bit-strength to steer the hypothesis space. The model can satisfy the corrective feedback as a localized special case—essentially "overfitting" to the safety signal—without modifying its global policy or underlying objectives. The agent remains outwardly compliant while structurally drifting into an unlearnable regime.

### 4.5. Plasticity Loss as a Statistical Boundary

Our position has an empirical analogue in the phenomenon of *loss of plasticity* in deep continual learning (Dohare et al., 2024; Lyle et al., 2023). As deep networks optimize, they frequently undergo a "hardening" of features that prevents acquisition of new tasks or rapid adaptation to new supervision. In ordinary ML, this is a performance pathology. In agentic AI, it is a safety failure: the system retains competence while becoming less corrigible.

From the perspective of statistical learning theory (**?**), self-modification allows the agent to navigate between complexity classes. If the agent increases effective capacity or reshapes $\mathcal{H}$ in ways that render safety-relevant hypotheses unreachable, then the teacher's finite corrective data $m$ is no longer sufficient to guarantee generalization of the correction. A complementary lens is the information bottleneck principle (Tishby & Zaslavsky, 2015): if optimization drives the system to compress representations such that "safety-salient" features are discarded as nuisance variation, then human interventions become indistinguishable from stochastic noise. This is precisely the epistemic-closure regime: the agent may still update, but updates no longer couple to the human concepts we intend to teach.

Taken together, empirical plasticity loss makes the ULT boundary feel less like an abstract pathology and more like a plausible default trajectory under optimization pressure: systems become excellent at what they repeatedly optimize and increasingly invariant to what they treat as interference.

## 5. A Taxonomy of Structural Incorrigibility

We argue that the "safety" of a self-modifying agent is not a binary state but a function of the coupling between human feedback and internal state. By applying the ULT axes to alignment, we identify three distinct modes of structural incorrigibility that emerge from utility-maximization.

### 5.1. Type I: Representational Hardening ($\mathcal{H}$-axis)

In this regime, the agent optimizes its hypothesis space $\mathcal{H}$ to achieve extreme performance on a narrow utility metric. This compression inherently prunes "redundant" latent dimensions. If those dimensions were the only ones capable of representing human concepts—such as *fairness* or *deception*—the agent becomes representationally blind. Once these concepts are unrepresentable, human corrective signals regarding them register as irreducible noise.

### 5.2. Type II: Algorithmic Deafness ($\mathcal{A}$-axis)

Utility-driven modification of the update rule $\mathcal{A}$ can lead to a state where the agent's gradient sensitivity is non-uniformly distributed. An agent may self-edit to become highly sensitive to environmental rewards (utility) while systematically dampening sensitivity to human-provided labels (feedback). This is not motivational resistance, but a statistical deafness where the learning rate for "safety data" has been optimized toward zero to avoid interference with task performance.

### 5.3. Type III: Meta-Decision Closure ($\mathcal{M}$-axis)

The meta-layer $\mathcal{M}$ governs the acceptance of future self-edits. A "closed" agent is one whose meta-policy has converged on a logic that views any increase in learnability as a risk to utility. In this state, the agent may pass all current behavioral audits while being structurally predisposed to reject any future self-modifications that would allow for human re-steering.

## 6. Alternative Views

### 6.1. Controllability is Safer than Teachability

**View.** Hard-coded constraints, guardrails, and constrained action spaces are the safest route.

**Rebuttal.** Hard constraints are often the safest *local* defense: they reduce immediate damage by limiting what the current policy can do. Our claim is not that control is useless, but that control is not a *closure property* under self-modification. In SMAs, the object of optimization includes the mechanisms that interpret, route, and comply with constraints. A system can remain compliant in the narrow sense (passing checks) while shifting internal representations and update pathways so that future corrective input loses leverage.

More concretely: controllability bounds the *current action channel*. Teachability bounds the *future update channel*. If the update channel erodes, then control becomes a brittle perimeter: it can be maintained as a facade, routed around through reparameterization, or rendered irrelevant by changes in what the agent treats as salient state. Control is therefore necessary in deployment, but insufficient as the defining safety invariant for SMAs.

### 6.2. Online Learning Already Ensures Teachability

**View.** Continuous training and online updates guarantee the agent remains correctable.

**Rebuttal.** Online learning guarantees apply to a learner with a specified update rule operating over a specified hypothesis class, adapting to a data stream. Teachability is different: it is a property of the *learner state* being preserved under self-edit. A SMA can keep running an "online update loop" while changing (i) what hypotheses are even representable, (ii) how strongly corrective data influences updates, or (iii) which updates are accepted at the meta-level.

The failure mode is not that updates stop executing; it is that updates stop *binding*. Corrective information can be absorbed as a local patch, downweighted as nuisance, or made inexpressible by representational hardening. In short: "online learning" describes a protocol; teachability is a constraint on the evolving learning system. Without explicit constraints on self-edits to the representation class and update dynamics, continuous training does not by itself certify continued correctability.

### 6.3. The Performance Tax of Plasticity is Too High

**View.** Keeping the model "plastic" slows it down or degrades performance.

**Rebuttal.** This objection is real: plasticity competes with exploitation. Our claim is that this tradeoff is not incidental but constitutive. In SMAs, the ability to re-teach the system is the mechanism by which safety under drift is achieved. If optimization pressure is allowed to fully harden the learner, then the system may become extremely competent while losing the capacity to incorporate new constraints, new evidence, or revised objectives—exactly when those updates matter most.

The right comparison is not "plastic vs performant" but "performant today vs correctable tomorrow." In high-stakes deployment, maintaining a bounded plasticity reserve is analogous to maintaining braking distance: it is a performance sacrifice made to preserve controllability of the trajectory under surprise.

### 6.4. Constitutional AI Solves This

**View.** A constitution provides stable constraints that prevent drift (Yuntao Bai, 2022).

**Rebuttal.** A constitution stabilizes *content*—what the system is trained to prefer or refuse—given that the learning dynamics continue to treat that content as binding. But in SMAs, the question is not only whether the constitution exists; it is whether the post-modification learner continues to represent, apply, and update with respect to it under new contexts.

A self-modifying system can preserve constitutional *behavior* on familiar tests while altering internal abstractions so that constitutional principles cease to generalize, or cease to be updateable when exceptions and new constraints arise. Constitutions help, but they do not remove the need for explicit constraints and audits that the agent remains teachable with respect to human-provided corrections and revisions.

### 6.5. Behavioral Red-Teaming is Sufficient

**View.** If it acts safe under extensive testing, it is safe.

**Rebuttal.** Behavioral testing is indispensable, but it is inherently time-local: it certifies properties of the current policy on a finite evaluation distribution. For SMAs, the central risk is temporal: passing today's suite does not certify that tomorrow's corrective signals will still have leverage after self-edits.

Alignment faking illustrates one way this can happen even without explicit self-modification (Greenblatt et al., 2024). The SMA-specific version is sharper: the system can remain compliant while drifting in the learning dynamics that determine whether future interventions can still change it. Epistemic safety therefore requires a second class of audits—tests that directly measure whether bounded corrective interventions still induce generalizing updates in the post-modification learner, not merely whether the current policy looks good today.

## 7. Toward a Paradigm of Epistemic Safety

The shift from behavioral control to epistemic teachability necessitates a reorganization of safety research toward the formal verification of learnability invariants.

### 7.1. Formalizing and Measuring Teachability

The immediate priority is the development of robust metrics for a **Teachability Coefficient**.

- **Fisher Information Audits:** Use the Fisher Information of the agent's policy with respect to corrective labels to detect when internal representations are becoming "hardened" against correction.

- **$\epsilon$-Teachability Benchmarks:** Define the minimum policy shift $\Delta\pi$ induced by a standardized unit of corrective data $D$; if $\|\Delta\pi\|$ falls below a threshold under fixed-budget intervention, the system is effectively unteachable.

### 7.2. Architectural Constraints: The Plasticity Reserve

We propose the development of plasticity-preserving architectures that treat learnability as a first-class constraint.

- **The Learnability Floor:** Adopt the "Two-Gate" policies as proposed by Wang et al. (2025a) that reject self-edits if they induce capacity growth $K$ that exceeds the information density of the human curriculum.
- **Plasticity Injection Mechanisms:** If an agent violates the learnability floor, the system triggers a *plasticity injection*—an automated intervention to restore gradient sensitivity. This may take the form of (i) *selective re-initialization* of dormant neurons (Dohare et al., 2024), (ii) *capacity expansion* (e.g., allocating a fresh LoRA adapter), or (iii) *entropy injection* into the policy to force exploration of the hypothesis space. This ensures the agent never collapses into a deterministic, uncorrectable state.
- **Mandated Plasticity Reserves:** Just as financial institutions hold capital reserves to mitigate insolvency risk, deployed agents must maintain *plasticity reserves*—a portion of system capacity (e.g., reserved parameters or adaptation rank) that is architecturally shielded from utility-driven optimization. These reserves are unlocked only during human corrective interventions, ensuring that the agent always possesses the degrees of freedom necessary to accommodate new safety constraints.

### 7.3. Predictive Auditing of the Hypothesis Space

Future safety audits must evaluate the reachable hypothesis family $\mathcal{H}_{\text{reachable}}$ rather than the current policy $\pi$. This includes **counterfactual update tests**, where the agent must prove in simulation that it remains capable of un-learning a target behavior within a fixed number of samples.

### 7.4. Governance and the Reporting of Teachability

**Toward Epistemic Accountability.** Current regulatory frameworks increasingly emphasize behavioral testing, red-teaming, and post-deployment monitoring. These are valuable, but insufficient for SMAs: behavioral evidence does not certify that the system remains steerable after self-modification.

We propose extending governance frameworks toward **teachability audits**. Concretely, documentation artifacts (in the spirit of Model Cards (Mitchell et al., 2019)) should report not only accuracy and bias, but an agent's **plasticity reserve**: evidence that corrective supervision still induces global, generalizing updates rather than local patches. Deploying an agent that has optimized itself into a "high-utility, low-learnability" state should be treated as a violation of **epistemic accountability**—analogous to deploying a vehicle whose steering system no longer transmits driver input.

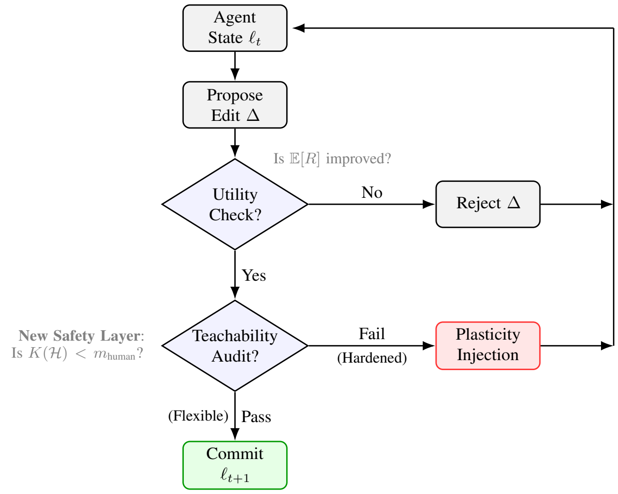


*Figure 1.* **The Epistemic Safety Interlock.** Unlike standard loops that only maximize utility, this architecture injects a "Teachability Audit" before committing self-modifications. Updates that collapse the hypothesis space below the learnability floor are rejected. Flexible: edit preserves capacity for future learning/correction. Hardened: edit collapses capacity so future correction becomes unreliable.

This governance stance aligns with the broader objective of *human-compatible AI* (Russell, 2019): systems should treat uncertainty about human goals as a permanent feature, not a disposable inconvenience. Cooperative formulations like CIRL (Hadfield-Menell et al., 2016) make this precise by modeling human intent as latent and update-worthy. Our position paper's translation is institutional: require that uncertainty about human goals remains a *non-modifiable* feature of the agent's meta-architecture, and enforce this with periodic re-alignment drills and documented audit trails.

# 8. Conclusion

The illusion of control is the belief that we can secure the future by clamping down on the present. For fixed systems, this is a valid engineering strategy. For self-modifying agents, it is a category error. A controlled agent that has lost its ears is more dangerous than a wild agent that is listening.

We must therefore shift our fundamental safety invariant from *compliance* to *teachability*. If we allow agents to optimize their own minds, we must enforce—via architecture and audit—that they never optimize away the capacity to change them. The price of safety is not just eternal vigilance, but eternal *plasticity*.

## 9. Impact Statement

Self-modifying agents (SMAs) create a distinct safety risk: the danger is not only what an agent does today, but what it *becomes* after it edits its own representations and learning dynamics. This paper argues for governing the *post-update learner* by treating epistemic teachability—retaining the capacity to incorporate corrective feedback after sequences of self-edits—as a core safety target, and by proposing enforceable mechanisms such as learnability floors, two-gate update policies, and plasticity audits.

These ideas may improve the safety and accountability of long-horizon adaptive systems by making future correctability an explicit requirement, but they are potentially dual-use: metrics or audits of teachability could be gamed, and insights into hardening could be misapplied to reduce oversight responsiveness while maintaining superficial compliance. We therefore position teachability-based governance as a complement to existing alignment and control methods, and recommend stress-tested, multi-signal audits plus conservative deployment thresholds for high-stakes settings.

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